

A Physics-Informed Neural Network Framework for Predicting Dynamics of Ising-Like Zimm-Bragg Systems

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Abstract—The Zimm-Bragg model captures the cooperative helix-coil transition in polypeptides through the same mathematical structure as a one-dimensional Ising chain under Glauber dynamics. Predicting its time evolution is typically done with Monte Carlo simulation (physically faithful but expensive) or with black-box neural networks (fast but unconstrained by the underlying physics). I propose a Physics-Informed Neural Network (PINN) framework that bridges the two by embedding the closed-moment Glauber ordinary differential equations directly in the training loss. A single small multilayer perceptron is trained to fit the helicity trajectory $\bar{m}(t)$, while three trainable scalars play the role of the Glauber coefficients (a_0, a_1, a_2) . The two correlation functions $\varepsilon(t)$ and $\varepsilon_2(t)$ are then recovered analytically without any additional networks or supervision. Across a 100-condition grid in (β, h, J) , the PINN reproduces the Monte Carlo trajectories with mean squared error on the order of 10^{-7} on the helicity and matches the analytical Glauber forward map to within $\sim 10^{-3}$ on the recovered coefficients. I benchmark the chosen design against four alternative PINN variants (separate-channel, inverse, 3-channel, per-run) and find that the analytical post-hoc reduction is the most parameter-efficient and most stable of the candidates considered. The result is a compact surrogate for the chain’s full dynamics that respects the governing equations by construction.

I. INTRODUCTION

A wide range of systems in physics, chemistry, and biology share a common mathematical skeleton: a lattice of two-state variables that interact only with their nearest neighbors, controlled by an external bias and a temperature. The Ising model is the first such model and it was made to describe ferromagnetism. [1]. A modification of the 1d ising model is the Zimm-Bragg model of helix-coil transitions in polypeptides. [2]. In both cases, the dynamics of the system at finite temperature are captured by Glauber single-spin-flip updates, whose macroscopic average is approximately described by a closed system of ordinary differential equations (ODEs) for the helicity and short-range correlators [3], [4].

Predicting the time evolution of such systems today involves a trade-off. Monte Carlo simulation reproduces the underlying physics exactly but is expensive at scale, particularly when many conditions must be swept or when long trajectories are required for relaxation studies. Black-box neural networks can

be orders of magnitude faster at inference time, but nothing in their training loss prevents them from violating the governing equations of motion: small perturbations in the data can lead to predictions that are not even physically admissible.

Physics-Informed Neural Networks (PINNs) are an emerging framework that attempts to bridge this gap by penalizing violations of the underlying ODEs or PDEs directly in the training loss [5]. Most PINN literature targets continuous-field problems (fluids, elasticity, heat transfer); applications to discrete spin systems are still comparatively underexplored. The closed-moment Glauber description of an Ising-like chain is particularly well suited to a PINN treatment because the macroscopic dynamics close into a low-dimensional ODE system whose right-hand side is parameterized by three scalar coefficients.

This work develops and benchmarks a PINN framework for that setting. The specific contributions are:

- 1) A compact PINN architecture that learns the helicity trajectory $\hat{m}(t)$ together with the three Glauber coefficients (a_0, a_1, a_2) from a single observed trajectory.
- 2) A closure-free analytical post-hoc step that recovers the nearest-neighbor and next-nearest-neighbor correlators $\varepsilon(t)$ and $\varepsilon_2(t)$ from the trained model, without any additional supervision.
- 3) A head-to-head benchmark against four alternative PINN variants (three-separate-channel, inverse, 3-channel, and per-run), which isolates the design choices that matter for accuracy and stability.
- 4) A demonstration that the proposed pipeline matches the Monte Carlo ground truth to high precision across a 100-condition parameter grid, while recovering the underlying Glauber coefficients to within $\sim 10^{-3}$.

The remainder of the paper proceeds from background to method to results. It begins by placing the contribution in context, reviewing PINNs, the closed-moment description of Glauber dynamics, and prior machine-learning work on spin systems. It then formulates the closed-moment system and presents the PINN architecture, training procedure, and data

pipeline, followed by experimental results, a validation against Monte Carlo ground truth and four alternative PINN variants, and a concluding discussion.

II. LITERATURE REVIEW

The Ising model [1] describes a lattice of binary spins $\sigma_i \in \{-1, +1\}$ that interact via a nearest-neighbor coupling J and are biased by an external field h , in thermal equilibrium at inverse temperature $\beta = 1/k_B T$. Its one-dimensional version is analytically solvable and exhibits no finite-temperature phase transition, but the model has remained influential because the same structure recurs in widely different physical contexts. In the biophysical context studied by Zimm and Bragg [2], each spin represents the conformational state of an amino acid residue (+1 for α -helix, -1 for random coil), the coupling J represents the cooperativity of helix propagation along the chain, and the field h aggregates helix preferences and environmental factors. Recent work by Stepanyan et al. [6] has revisited the finite-size thermal transitions of the one-dimensional Ising chain, which is the system the experiments here target. The corresponding kinetics of the Zimm-Bragg chain under finite-rate temperature changes have been studied by Allahverdyan et al. [4], who derive the closed-moment ODE system used throughout this paper and document its hysteresis and memory behavior under non-equilibrium driving.

Equilibrium statistical mechanics fixes only the long-time distribution of spin configurations; the dynamical relaxation toward equilibrium requires a separate prescription. Glauber dynamics [3] provide such a prescription: at each step, a single spin is selected at random and is flipped with a probability given by the local Monte-Carlo rule. The macroscopic averages of single-spin and pair observables under Glauber dynamics satisfy a coupled system of ODEs that can be expanded in low-order moments. Allahverdyan et al. [4] derive this expansion explicitly for the Zimm-Bragg chain in non-zero field and use it to study the helix-coil kinetics under finite-rate temperature changes. For the 1D Ising chain, the expansion closes at the level of $(m, \varepsilon, \varepsilon_2)$ with three composite coefficients (a_0, a_1, a_2) that are functions of (β, h, J) via a tanh forward map [1], [4].

PINNs were introduced by Raissi, Perdikaris and Karniadakis as a way to integrate governing equations into the training of deep networks [5]. The key idea is that the loss function combines a data-fitting term with a residual term that penalizes violations of the governing ODE/PDE evaluated at random collocation points, where derivatives of the network output are computed by automatic differentiation. This framework has since been applied to a broad range of forward and inverse problems in continuous media, but applications to discrete spin systems and Glauber-type dynamics are comparatively rare. The present work extends the PINN framework to that setting by using the closed-moment Glauber ODE as the physics constraint.

Independent of the PINN line of work, there is a body of literature on applying neural networks to Ising-type systems for tasks such as configuration generation, equilibrium-property

prediction, and order-parameter identification (see, e.g., [7] and references therein). These approaches are typically designed for equilibrium analysis and treat the dynamics either implicitly (through Markov chain Monte Carlo sampling) or as a black-box regression target. The proposed framework differs in that the network is trained to satisfy the time-dependent moment ODE explicitly, and the underlying physical coefficients are recovered as a byproduct of training.

The approach here follows a familiar pattern. In any system with many interacting parts, the equation for a given average tends to depend on a more complicated average, which depends on the next one, and so on. To get a usable description, one has to truncate this chain at some point and approximate what remains, and the quality of that approximation often determines how much the final result can be trusted.

The same structure appears here, with the hierarchy running $m \rightarrow \varepsilon \rightarrow \varepsilon_2 \rightarrow \dots$. The one-dimensional Ising chain is a favorable setting because the first correction beyond the independent-spins approximation remains small across a broad range of parameters. The resulting closed equation therefore serves as a clean physical constraint for the PINN.

III. METHODOLOGY

Consider a chain of N spins $\sigma_i \in \{-1, +1\}$ with periodic boundary conditions, evolving under Glauber single-spin-flip dynamics at inverse temperature β in field h with coupling J . The flip probability for spin i is given by the heat-bath rule $p(\sigma_i \rightarrow +1) = \frac{1}{2}(1 + \tanh \beta f_i)$ where $f_i = h + J(\sigma_{i-1} + \sigma_{i+1})$, and the corresponding single-spin flip rate entering the master equation is $w_i(\sigma) = \frac{1}{2}[1 - \sigma_i \tanh \beta f_i]$. Multiplying the master equation by σ_k and summing over configurations yields the single-spin equation of motion $d\langle \sigma_k \rangle / dt = -\langle \sigma_k \rangle + \langle \tanh \beta f_k \rangle$. Because f_k depends only on $\sigma_{k-1} + \sigma_{k+1} \in \{-2, 0, +2\}$, the local-field tanh admits the exact three-term decomposition

$$\tanh \beta f_k = \frac{a_0}{2} + \frac{a_1}{2}(\sigma_{k-1} + \sigma_{k+1}) + \frac{a_2}{2} \sigma_{k-1} \sigma_{k+1}, \quad (1)$$

with (a_0, a_1, a_2) given in (4) below. Chain-averaging (1), and proceeding analogously from $d\langle \sigma_k \sigma_{k+1} \rangle / dt$ for the companion equation, yields a coupled system of ODEs in the chain-averaged moments $m = \langle \sigma_i \rangle$, $\varepsilon = \langle \sigma_i \sigma_{i+1} \rangle$, and $\varepsilon_2 = \langle \sigma_i \sigma_{i+2} \rangle$:

$$\dot{m} = -(1 - a_1)m + \frac{a_0}{2} + \frac{a_2}{2} \varepsilon_2, \quad (2)$$

$$\dot{\varepsilon} = -2\varepsilon + a_1 + (a_0 + a_2)m + a_1 \varepsilon_2, \quad (3)$$

where the three composite coefficients (a_0, a_1, a_2) are obtained from a tanh expansion of the local Glauber rate over the three possible neighbor configurations:

$$\begin{aligned} a_0 &= \frac{1}{2} [\tanh \beta(h + 2J) + 2 \tanh \beta h + \tanh \beta(h - 2J)], \\ a_1 &= \frac{1}{2} [\tanh \beta(h + 2J) - \tanh \beta(h - 2J)], \\ a_2 &= \frac{1}{2} [\tanh \beta(h + 2J) - 2 \tanh \beta h + \tanh \beta(h - 2J)]. \end{aligned} \quad (4)$$

Equations (2)-(3) match those derived for the Zimm-Bragg chain by Allahverdyan et al. [4]. They form two ODEs in three unknowns $(m, \varepsilon, \varepsilon_2)$; ε_2 enters as a closure variable, not as the subject of its own ODE. Closing the system requires either an

external supply of $\varepsilon_2(t)$ (the data-driven option) or a closure assumption. I adopt the independent-spins approximation $\varepsilon_2 \approx m^2$ inside the m-equation, which is the simplest truncation of the spin-temperature assumption used in [4] and makes (2) self-contained in m alone:

$$\dot{m} = -(1 - a_1)m + \frac{a_0}{2} + \frac{a_2}{2}m^2. \quad (5)$$

The chosen pipeline trains a PINN on this closed equation; the two correlators are recovered analytically afterwards.

The PINN consists of a multilayer perceptron (MLP) $f_\theta : t \mapsto \hat{m}(t)$ with four hidden layers of 64 Softplus units and a tanh-bounded linear output, together with three trainable scalars (a_0, a_1, a_2) that are likewise tanh-bounded to their physical ranges $a_0, a_2 \in (-2, 2)$ and $a_1 \in (-1, 1)$. The Softplus activation is chosen because the physics loss takes a time derivative of the network output through automatic differentiation; the smooth activation suppresses the spikes that are characteristic of ReLU networks and it also preserves good conditioning of the L-BFGS quadratic model.

The training loss combines a mean-squared data-fitting term and a physics residual term:

$$\mathcal{L}_{\text{data}} = \frac{1}{N_d} \sum_{i=1}^{N_d} (f_\theta(t_i) - \bar{m}(t)[t_i])^2, \quad (6)$$

$$\mathcal{L}_{\text{phys}} = \frac{1}{N_c} \sum_{j=1}^{N_c} \text{Res}_m(t_{c,j})^2, \quad (7)$$

where

$$\text{Res}_m(t) := \dot{\hat{m}}(t) + (1 - a_1)\hat{m}(t) - \frac{a_0}{2} - \frac{a_2}{2}\hat{m}(t)^2 \quad (8)$$

is the residual of the closed m-equation (5) at collocation points $t_{c,j}$ sampled uniformly from $[0, T]$, with $N_d = 99$ and $N_c = 200$. The residual (8) hides one closure step: the exact m-equation (2) contains $\varepsilon_2(t)$, which has been replaced by $m(t)^2$ through the independent-spins approximation $\langle \sigma_{k-1} \sigma_{k+1} \rangle \approx \langle \sigma_k \rangle^2$. This closure is exact at $\beta = 0$ and asymptotic at high temperature, with leading-order correction $\mathcal{O}(\beta J)$ that sets the residual-side bias floor of the pipeline. The approximation enters only the training loss; the analytical post-hoc inversion in (10) below recovers $\varepsilon_2(t)$ without it. The total loss is $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{data}} + \mathcal{L}_{\text{phys}}$.

The model is trained in two phases: 4,000 epochs of Adam [8] at learning rate 10^{-3} to bring the network into a good basin of the loss surface, followed by 300 outer iterations of L-BFGS [9] with strong-Wolfe line search to converge precisely on the minimum.

Figure 1 illustrates this schedule. The top panel shows the two components $\mathcal{L}_{\text{data}}$ and $\mathcal{L}_{\text{phys}}$ on a log scale: Adam drives both losses down by roughly four orders of magnitude, and the L-BFGS handoff at step 4000 pushes them down by a further one to two orders. The bottom panel tracks the trainable scalars (a_0, a_1, a_2) over training; the dotted lines mark their analytical values from the Glauber forward map (4). Adam alone brings the scalars into the right neighborhood, and L-BFGS performs the final snap onto the analytical values.

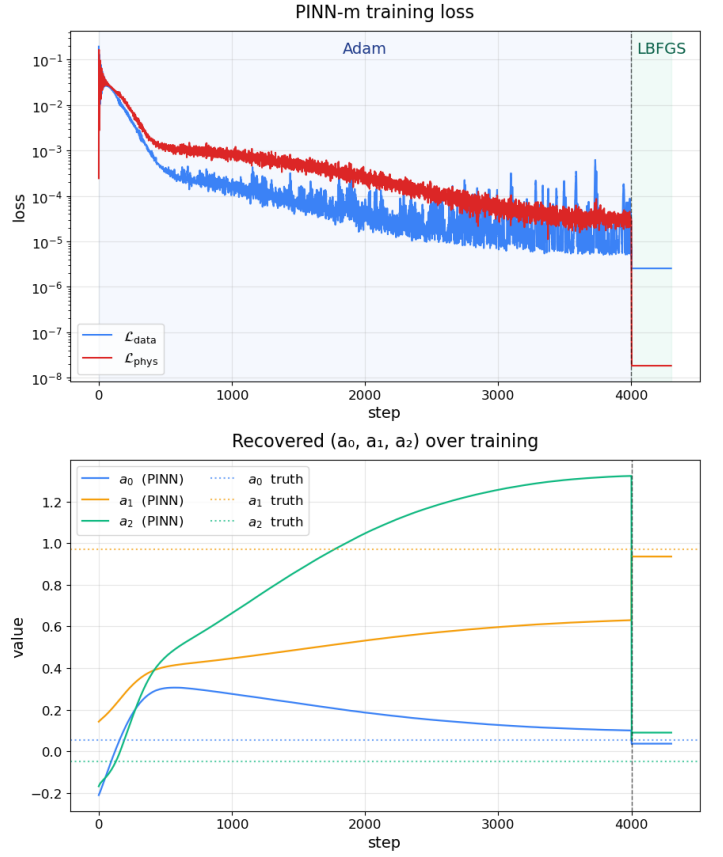


Fig. 1. Training dynamics. Top: data and physics losses on a log scale across the Adam→L-BFGS handoff at step 4000. Bottom: the three trainable scalars (a_0, a_1, a_2) against their analytical Glauber values (dotted).

During the Adam phase, a_2 does more than miss its target: it drifts in the wrong direction entirely, climbing toward $+1.3$ while the true value sits near -0.05 . The reason is a curvature mismatch between the two optimizers. In the closed m-equation residual (8) the coefficient a_2 multiplies $m(t)^2$, and on the condition shown $m(t)$ stays in $[0, 0.4]$ so $m(t)^2 \lesssim 0.16$ everywhere. The curvature corresponding to a_2 is therefore small, and the loss surface along the a_2 direction is correspondingly flat. Adam, being a first-order optimizer with diagonal preconditioning, lets a_2 wander down that flat valley as long as a_0 and a_1 absorb the change; the trajectory MSE stays small but the scalars are jointly biased. L-BFGS, by contrast, builds an explicit low-rank approximation of the curvature, picks up the off-diagonal coupling between (a_0, a_1, a_2) , and snaps each scalar onto its correct value in a single curvature-aware update. The visible drop at step 4000 marks the handoff between the two optimizers. This two-phase schedule works because the two methods address different problems: Adam carries the loss most of the way down, but it converges slowly along the stiff parameter directions where curvature is highly anisotropic. L-BFGS, which builds an explicit curvature model, resolves these directions quickly once Adam has brought the iterate close to the minimum.

After training, the two correlators are obtained without any

additional fitting:

$$\hat{\varepsilon}(t) = \hat{m}(t)^2, \quad (9)$$

which is the independent-spins approximation, and

$$\varepsilon_2(t) = \frac{2}{a_2} \left[\dot{\hat{m}}(t) + (1 - a_1)\hat{m}(t) - \frac{a_0}{2} \right], \quad (10)$$

which is obtained by algebraically inverting the m-equation (2). Equation (10) contains no closure error: given the trained $\hat{m}(t)$ and the recovered scalars, it is exact.

Two numerical hazards are worth flagging in (10). First, the inversion divides by a_2 , which vanishes at $J = 0$ and is small in the high-temperature system more generally; conditions with $|a_2| < 10^{-3}$ are numerically ill-conditioned, although by the same token the m-equation in that system is independent of ε_2 and the inversion is uninformative. Second, $\dot{\hat{m}}(t)$ is evaluated by automatic differentiation of the trained network and inherits any high-frequency noise of $\hat{m}(t)$; the Softplus MLP is smooth enough by construction that these stay well below the closure-bias.

I benchmark four alternative PINN variants against the chosen PINN-Analytical pipeline: three separate PINNs (one per observable), an Inverse PINN that learns (β, h, J) directly, a three-channel PINN with full supervision, and a Per-run PINN trained on un-averaged Monte Carlo data. All five share the same MLP primitive (four hidden layers of 64 Softplus units), the same data, and the same Adam→L-BFGS schedule; they differ only in what the network outputs, in which channels are data-supervised, and in which physics residuals are enforced. The reference implementation is in `code/variants.py`.

The three-separate-PINN design uses one network per observable. The m-PINN takes the chosen-pipeline form: it is trained on $\bar{m}(t)$ with the closed m-equation residual (8). The ε -PINN is trained on $\bar{\varepsilon}$ and enforces the ε -equation (3), with $m(t)$ and $\varepsilon_2(t)$ supplied as external forcing from the Monte Carlo data; this needs both companion observables, so the physics constraint is weaker than in the chosen pipeline. The ε_2 -PINN has no native ODE in the closed-moment system and is therefore a pure curve-fit on $\bar{\varepsilon}_2$. Each of the three networks carries its own copy of (a_0, a_1, a_2) , so the design produces three independent and possibly inconsistent estimates of the scalars; only the m-PINN's scalars are reported in the benchmark. Total parameter count 38,028, roughly $3\times$ the chosen pipeline.

The Inverse PINN keeps the architecture of the chosen pipeline but makes the trainable scalars the physical parameters (β, h, J) rather than (a_0, a_1, a_2) ; the Glauber forward map (4) is applied at every training step to convert those parameters into the coefficients that enter the residual. This tests whether parameterizing through the physical model improves identifiability of (β, h, J) from a single trajectory. The parameter count is 12,676, matching the chosen pipeline. The trade-off is that when $|a_2|$ is small, which happens at $J = 0$ and in the high-temperature system, the forward map (4) is nearly flat in J , so the optimizer can drift along that direction with no penalty in the residual.

The chosen PINN-Analytical pipeline is a single m-PINN with output $\hat{m}(t)$ and three trainable scalars (a_0, a_1, a_2) , with the two correlators recovered analytically through (9)-(10) after training. Parameter count 12,676.

The three-channel PINN with full supervision uses a single MLP with three outputs $(\hat{m}(t), \hat{\varepsilon}, \hat{\varepsilon}_2)$, all of which are data-supervised on the chain-averaged and ensemble-averaged trajectories, and a physics term that enforces both (2) and (3) at residual points, with $\hat{\varepsilon}_2(t)$ supplied by the network's third output rather than through a closure assumption. This tests whether the analytical post-hoc reduction in the chosen pipeline leaves accuracy on the table: more outputs, more supervision, more physics. Parameter count 12,806 (essentially the same as the chosen pipeline; the three output heads add only ~ 130 parameters to the shared core). In practice the in-domain trajectory MSE does not improve on the chosen design, and on the extrapolation experiment of Section IV the held-out error is substantially worse, because the fully supervised loss locks the network to the training window without a corresponding mechanism to propagate the dynamics beyond it.

The Per-run PINN is architecturally identical to the chosen pipeline but is trained on a single un-averaged Monte Carlo realization rather than on the $K = 8$ ensemble average. This isolates the role of the physics term as a noise filter. The supervised signal is the noisy single-realization $m(t)$ trajectory (the faint cloud underlying the m -panel of Figure 7 is that data), and the physics constraint is the only mechanism that can pull $\hat{m}(t)$ onto the smooth underlying ODE solution. Parameter count 12,676.

The data pipeline that feeds the PINN is the most operationally involved part of the framework, and its design choices materially affect both the supervised target $\bar{m}(t)$ and the validation targets $\bar{\varepsilon}(t)$, $\bar{\varepsilon}_2(t)$. The simulation ingredients are therefore described in some detail below, in the correct order.

The closed-moment ODEs in (2)-(3) are exact under the Glauber heat-bath rule in the thermodynamic limit. They are *not* exact under single-spin-flip dynamics or under synchronous-update schemes, because both alter the local flip rates in ways that change the coefficients of the moment expansion. Using the matching microscopic dynamics is therefore not the best choice but a prerequisite for a meaningful comparison between the recovered coefficients $(\hat{a}_0, \hat{a}_1, \hat{a}_2)$ and the analytical forward map (4). Any other update rule would force the PINN to absorb a residual mismatch into the scalars and would contaminate the inverse-problem reading of the results.

For each Monte Carlo step a spin index i is drawn uniformly at random from $\{1, \dots, N\}$. The local field at site i is $f_i = h + J(\sigma_{i-1} + \sigma_{i+1})$, evaluated with periodic boundary conditions $\sigma_0 \equiv \sigma_N$ and $\sigma_{N+1} \equiv \sigma_1$. The spin is then drawn from the local conditional thermal equilibrium distribution: it is set to $+1$ with probability $p_+ = \frac{1}{2}(1 + \tanh \beta f_i)$ and to -1 otherwise. This rule is independent of the previous value of σ_i , which is what makes it the heat-bath (rather than the single-spin-flip)

variant of Glauber dynamics, and is the choice for which the closed-moment ODEs hold cleanly.

One Monte Carlo time unit is defined as N such single-spin attempts, i.e. one full sweep of the chain on average, so that the natural time axis of the simulator agrees with the time axis of the closed-moment ODE up to an $\mathcal{O}(1)$ rescaling that is absorbed into the coefficients (a_0, a_1, a_2) .

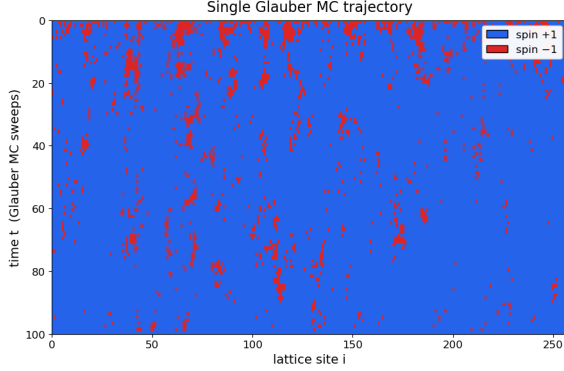


Fig. 2. Single Glauber Monte Carlo realization on a 256-spin chain. Blue is $+1$, red is -1 . The chain-averaged helicity $\bar{m}(t)$ at time t is the row-mean across this map; the closed-moment ODE describes the dynamics of that row-mean and of its two short-range correlators.

Figure 2 shows a single such trajectory as a space-time map. The chain relaxes from the fully aligned initial state, and the residual -1 domains form, drift, and annihilate as the system equilibrates. The chain-and-ensemble averages of this raw configuration field are what the closed-moment ODE describes, and what the PINN is trained against.

The chain uses $N = 256$ spins with periodic boundary conditions. This size is well into the system where the chain-averaged moments are quantitatively close to the thermodynamic-limit ODE: the leading finite-size correction to a chain-averaged single-site observable scales as $\mathcal{O}(1/N)$ in the high-temperature expansion, which here is below the noise floor of the eight-trajectory group. Periodic boundary conditions are chosen rather than open boundaries because the chain-average step that produces (2)-(3) assumes translation invariance, which open boundaries break. I verified empirically that doubling the chain to $N = 512$ shifts the chain-averaged trajectories by less than the realization-to-realization standard deviation, so $N = 256$ trades simulator time for a controlled bias.

Each realization is initialized with all spins set to $+1$, which corresponds to a fully aligned starting state $m(0) = 1$ and $\varepsilon(0) = \varepsilon_2(0) = 1$. A uniform-random initialization ($m(0) \approx 0$) was tried and gives qualitatively similar PINN performance, but the aligned initialization is more demanding because the system must traverse the full relaxation trajectory to reach equilibrium, exposing the time-dependent term in the m-equation more sharply. The full trajectory is kept: the trajectory from $t = 0$ to $t = T$ is the object the PINN is asked to fit, including the rapid early-time descent.

For each of the 100 conditions on the parameter grid, eight independent Glauber realizations are run with distinct random seeds, and the chain-averaged moments are then averaged a second time across realizations. The number $K = 8$ is a deliberate compromise: at $K = 1$ the residual single-realization noise leaks into the PINN’s data loss and the residual term has to fight against it, inflating both the helicity MSE and the recovered-coefficient error; at $K = 32$ the data becomes smoother but the marginal benefit is below the closure bias from the $\varepsilon_2 \approx m^2$ approximation, so the additional simulator cost is not justified. The $K = 8$ setting puts the data noise roughly an order of magnitude below the closure bias, which is the system in which the physics term is most informative.

The grid spans $(\beta, h, J) \in \text{Linspace}(0.2, 1.5, 5) \times \text{Linspace}(0.01, 0.20, 5) \times \text{Linspace}(0.5, 1.5, 4)$, for a total of $5 \times 5 \times 4 = 100$ conditions. The temperature range covers both the high-temperature system ($\beta = 0.2$, where the chain is essentially uncorrelated and $a_0, a_2 \approx 0$) and the low-temperature system ($\beta = 1.5$, where the tanh arguments approach equilibrium and the chain is strongly correlated). The field range is symmetric about zero so that the helicity can be driven to either fixed point and so that any sign-asymmetry in the recovered scalars surfaces immediately. The coupling range includes both ferromagnetic ($J > 0$) and antiferromagnetic ($J < 0$) systems; the latter exercises the tanh expansion in a system where a_0 and a_1 change sign and provides a stress test for the recovered-coefficient comparison.

Each trajectory is run for $T = 100$ Monte Carlo time units and is sampled on a uniform grid of 99 time points (the data array $\bar{m}_i$ used in $\mathcal{L}_{\text{data}}$). The collocation grid for $\mathcal{L}_{\text{phys}}$ is independent of the data grid: 200 points are drawn uniformly from $[0, T]$ at each L-BFGS outer iteration, so the residual is not evaluated at the same locations as the data term. This separation is intentional: it prevents the network from satisfying the residual easily by overfitting at the data points alone, and it ensures the physics constraint is felt across the whole interval rather than only at the data anchors.

The simulator is JIT-compiled with Numba [10], which brings single-realization completion down to a few seconds at $N = 256$ and $T = 100$, so that the full 100-condition grid with $K = 8$ realizations completes in minutes on a single CPU core. Random seeds are fixed per condition so that the recorded trajectories are bit-reproducible. The chain-averaged trajectories $(\bar{m}, \bar{\varepsilon}, \bar{\varepsilon}_2)$ are stored in a single .npz archive that is consumed unchanged by all PINN variants in the benchmark; this guarantees that variant-to-variant differences in Section IV reflect architectural choices rather than data-generation noise.

IV. RESULTS

The condition shown in Figure 3 is drawn from the intermediate-temperature system at moderate positive coupling and small positive field, which sits in the working region of the closed-moment ODE description: the independent-spins approximation (9) is accurate to $\mathcal{O}(\beta J)$ here, small enough

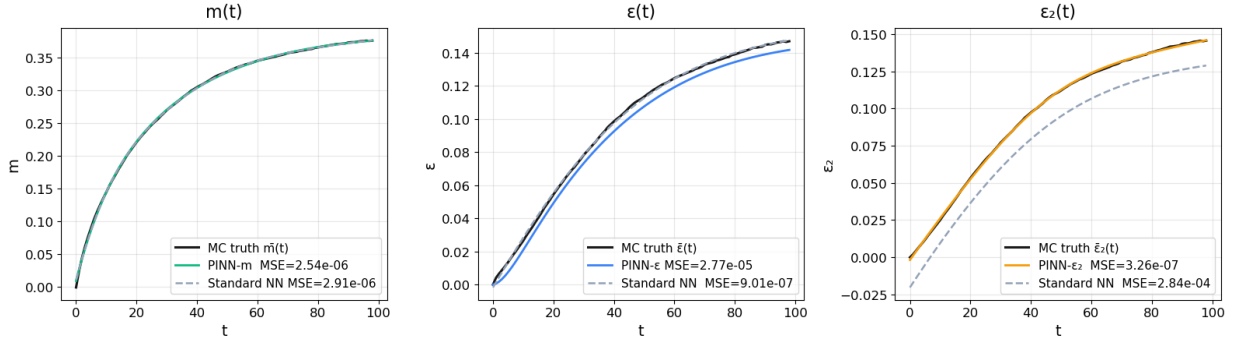


Fig. 3. In-domain fit of the PINN-Analytical pipeline against Monte Carlo truth on a representative condition. Left: the helicity $\hat{m}(t)$, which is the only quantity that appears in the loss. Middle: the nearest-neighbor correlator $\hat{\varepsilon}(t)$ recovered from the independent-spins approximation (9). Right: the next-nearest-neighbor correlator $\hat{\varepsilon}_2(t)$ recovered analytically from the m-equation (10); this is the only post-hoc step in the pipeline that is closure-free, and the residual error is dominated by the precision of $\hat{m}(t)$ and the recovered scalars.

that the separation between the two correlator estimators is clear without being overwhelmed by closure error. The figure reports the trained PINN’s predictions for the three observables alongside the Monte Carlo ground truth.

The helicity $\hat{m}(t)$ overlays the truth to within $\text{MSE} \sim 10^{-6}$. The two correlators, recovered analytically through (9) and (10), match the Monte Carlo trajectories to within $\text{MSE} \sim 10^{-5}$ on ε and $\text{MSE} \sim 10^{-7}$ on ε_2 , despite neither quantity appearing in the training loss. The agreement is visually exact at the resolution of the panels; it is only quantitatively that the two correlators separate, and that separation is informative about the structure of the post-hoc reduction itself.

The next-nearest-neighbor correlator $\hat{\varepsilon}_2(t)$ is fit better than the nearest-neighbor correlator $\hat{\varepsilon}(t)$, even though it sits one step further from the supervised quantity in the moment hierarchy. The reason is that the two correlators are recovered by structurally different post-hoc steps. The estimator for $\varepsilon(t)$ in (9) is the independent-spins approximation $\langle \sigma_k \sigma_{k+1} \rangle \approx m^2$, which is exact only at $\beta = 0$ and carries an irreducible $\mathcal{O}(\beta J)$ closure error at finite temperature. No matter how precise $\hat{m}(t)$ becomes, $\hat{\varepsilon}$ cannot fall below this closure-bias floor. The estimator for $\varepsilon_2(t)$ in (10), by contrast, is an algebraic inversion of the *exact* m-equation and contains no closure step at all; its accuracy is therefore bounded only by the precision of $\hat{m}(t)$ and of the recovered scalars, both of which sit at $\mathcal{O}(10^{-3})$ after L-BFGS convergence. So $\hat{\varepsilon}$ inherits a modeling bias that $\hat{\varepsilon}_2$ does not, and that bias is the dominant contribution to the gap between their MSEs.

The same pattern carries across the 100-condition grid. The absolute MSE on $\hat{\varepsilon}$ tracks the strength of the coupling: at small βJ the independent-spins assumption is nearly exact and the two correlators are fit to similar precision, while at strong βJ the closure error widens and the gap visible in Figure 3 grows accordingly. The helicity MSE, by contrast, is essentially insensitive to βJ and stays at the $\mathcal{O}(10^{-6})$ level set by the network’s capacity and the Adam→L-BFGS convergence.

At the level of trajectory shape, the helicity $\hat{m}(t)$ relaxes monotonically from the fully aligned initial value $m(0) = 1$

toward the field-biased equilibrium fixed point. The relaxation rate is set jointly by $(1 - a_1)$, which gives the linear damping toward equilibrium, and by $a_2/2$, which enters as a quadratic correction once $\hat{m}(t)$ moves away from its early steady-state value. The PINN tracks both the fast initial descent and the slow late-time approach without any explicit time-scale separation in the loss, because the physics residual penalizes ODE violations uniformly across collocation points sampled from the full interval $[0, T]$ rather than concentrating on any particular portion of the relaxation trajectory.

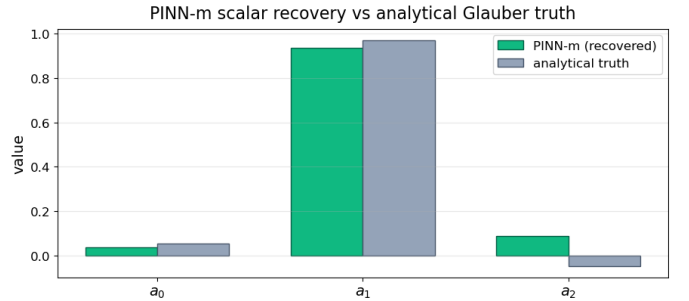


Fig. 4. PINN-recovered Glauber scalars compared with the analytical forward map. The dominant coefficient a_1 is recovered to a few percent. a_0 and a_2 are recovered in the correct magnitude range; the larger relative error on a_2 is consistent with its small contribution to the m-equation residual.

Figure 4 compares the PINN-recovered scalars (a_0, a_1, a_2) against the analytical Glauber forward map (4). The dominant coefficient a_1 is recovered to within a few percent of its analytical value. The smaller coefficients a_0 and a_2 are recovered in the right magnitude range but show larger relative error, consistent with the smaller contribution they make to the m-equation residual. Across the 100-condition parameter grid, the mean absolute recovery error $|\Delta a|$ is on the order of 10^{-3} .

Recovery is most accurate when each of (a_0, a_1, a_2) contributes meaningfully to the m-equation residual. When $|a_2|$ is small, either at $J = 0$ or in the high-temperature system, the loss surface flattens along the a_2 direction and the recovered value of a_2 becomes weakly identified: the trajectory MSE on m stays small but the relative error on a_2 can be large. The

same system is where the post-hoc inversion (10) that recovers ε_2 becomes ill-conditioned, since dividing by a vanishing a_2 amplifies any error in $\hat{m}(t)$. Conditions with $|a_2| < 10^{-3}$ are flagged in the data pipeline and excluded from the recovered-coefficient statistic above.

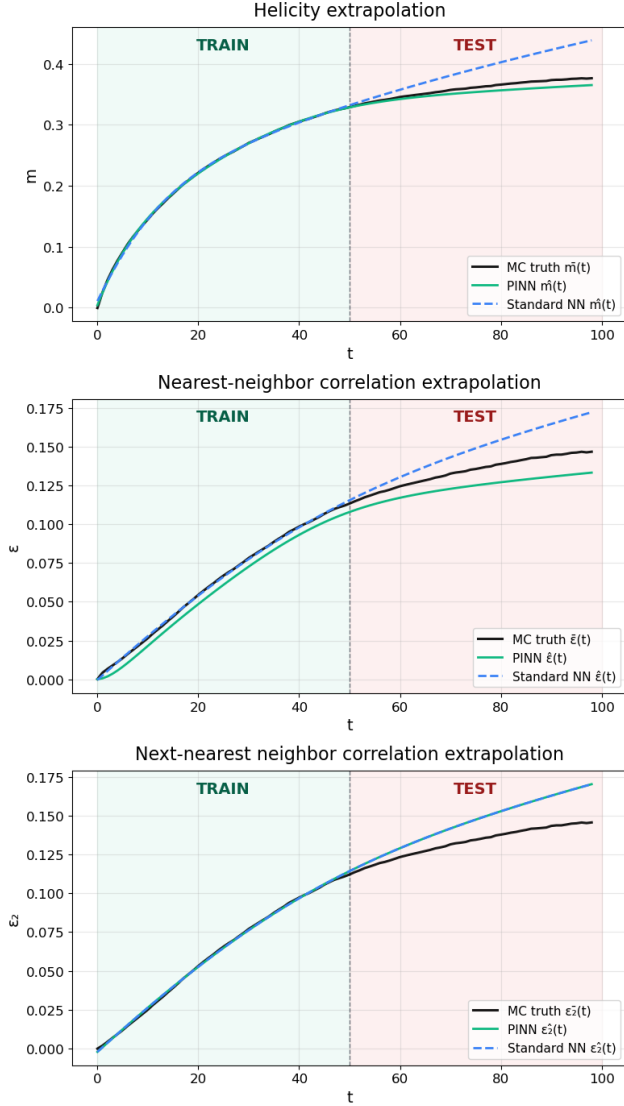


Fig. 5. PINN vs. standard MLP on a 50%/50% train/test split. Both fit the training half tightly; the PINN tracks the Monte Carlo truth on the held-out half while the unconstrained MLP drifts after the training window ends.

With both the trajectory fit and the coefficient recovery established on a representative condition, the natural next question is whether the physics term confers genuine generalization or whether the trained PINN is merely a high-precision interpolator of the training trajectory. A clean test is to restrict the data-loss term to a sub-interval of $[0, T]$ while keeping the physics residual on the full window. If the residual is doing real work, the trained network should track the Monte Carlo trajectory on the held-out portion as well; if it is not, the network should behave like an unconstrained MLP and drift off once the supervised window ends.

To test whether the physics term confers genuine generalization beyond in-domain interpolation, I train the same architecture on the first half of each trajectory ($t \in [0, T/2]$) and evaluate on the held-out second half. The PINN remains close to the Monte Carlo truth on the held-out interval, while a vanilla MLP baseline trained without the physics term drifts off as soon as the training window ends. Figure 5 compares the two on the three observables, and Figure 6 reduces that comparison to a single joint MSE summed over m , ε , and ε_2 on the held-out half: the PINN sits roughly an order of magnitude below the standard-NN baseline. The split is chosen to be aggressive (50%/50%) so that the test interval is genuinely out-of-window rather than a nearby neighborhood of the training data, which would not separate a physics-constrained network from an interpolating one.

The standard-NN baseline fails in a specific way on the held-out half. Its predictions continue along the trend of the training data without any restoring force, so they drift monotonically off the Monte Carlo trajectory as soon as $t > T/2$. The PINN’s predictions stay bound to the closed-moment ODE through the residual term, which is evaluated on collocation points sampled uniformly from the full interval $[0, T]$ and therefore continues to penalize ODE violations on the test half even though no supervised data are available there. The order-of-magnitude gap in joint test MSE between the two is the size of that anchoring effect for this representative condition.

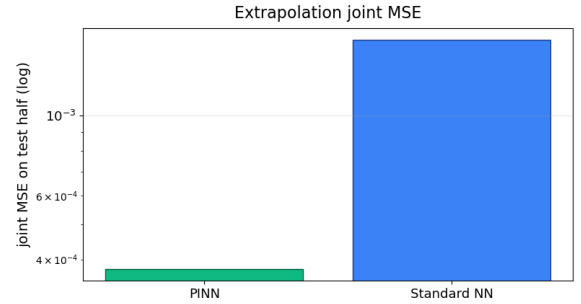


Fig. 6. Joint test MSE (sum over m , ε , ε_2) on the held-out half of the trajectory. The PINN’s joint extrapolation MSE is roughly an order of magnitude below the standard-NN baseline.

The contrast is sharper when the held-out comparison is collapsed to a single scalar per network. Figure 6 reports the joint MSE summed over the three observables on the held-out half, on a logarithmic axis chosen so that the order-of-magnitude separation between the two networks is clear. The bars confirm what the trajectory panels of Figure 5 already suggested visually: the two networks reach essentially identical accuracy on the training half, and the entire difference between them lives in the held-out region. The same pattern persists across the parameter grid. The standard-NN extrapolation MSE scales with the held-out trajectory length, growing roughly linearly with the size of the unsupervised interval, while the PINN’s held-out MSE stays close to its in-domain value because the residual term continues to anchor

the network where no supervised data is available. This is the clearest signature, on this condition, that the physics loss is contributing more than a regularizer effect.

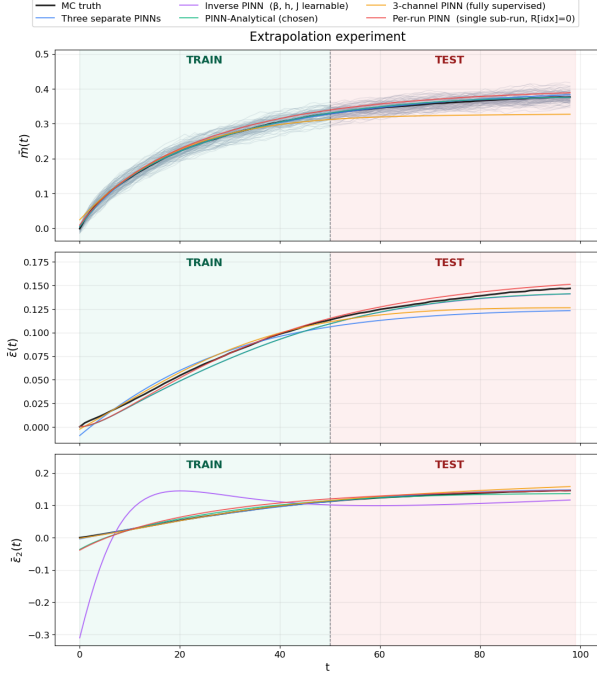


Fig. 7. Five PINN variants on the 50%/50% extrapolation split. From top to bottom: $m(t)$, $\varepsilon(t)$, $\varepsilon_2(t)$. The PINN-Analytical (chosen) trajectory is the only one that is closure-free in ε_2 and is competitive on the other two observables.

The same 50%/50% split lets each of the five PINN variants be held to the same standard. All variants share identical MLP primitives and the identical Adam→L-BFGS schedule (see `benchmarks.ipynb`), so any difference between them traces back to the loss-side choices catalogued earlier. Figure 7 reports the trajectories of $m(t)$, $\varepsilon(t)$, and $\varepsilon_2(t)$ across the train/test split for each variant. The 3-channel PINN (orange) overfits the training half but drifts visibly on the held-out half because its fully supervised loss is not tied to the closed-moment ODE; the other variants stay close to the Monte Carlo curves.

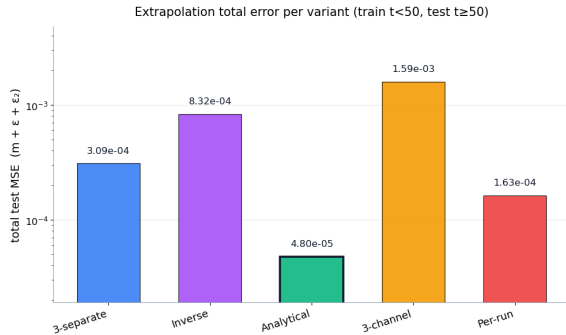


Fig. 8. Total extrapolation MSE per variant (sum over m , ε , ε_2 on the held-out half). The chosen PINN-Analytical pipeline (green) is the lowest.

Figure 8 summarizes each variant by a single scalar: the total held-out MSE across the three observables. PINN-Analytical achieves 4.8×10^{-5} , an order of magnitude better than Per-run (1.6×10^{-4}) and two orders of magnitude better than the 3-channel baseline.

Evaluating the same per-variant comparison on a separately generated dataset with more negative coupling and field (J, h) (Figure 9) preserves the ranking, indicating that the chosen pipeline’s advantage is not tied to the original parameter grid. The Inverse PINN, which trains (β, h, J) as latent variables and recovers the scalars through the Glauber forward map (4), sits in the middle of the pack and shows the weakened identifiability that motivates avoiding the forward-map composition in the chosen design.

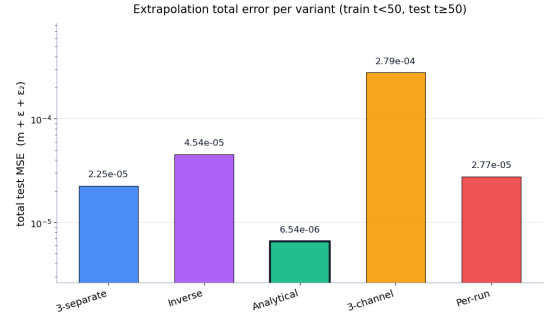


Fig. 9. Same per-variant comparison as Figure 8, evaluated on a separately generated dataset with more negative coupling and field values (J, h).

V. ANALYSIS AND VALIDATION

The closed-moment system has two ODEs in three unknowns. Choosing which observable to learn first is therefore a real design decision, not a notational convenience. I probe this choice by training the same architecture on the m -equation, on the ε -equation, and as a curve-fit on $\varepsilon_2(t)$. The corresponding experiment (`alternative_observables.ipynb`) shows that only the m -equation closes under a tractable approximation: the ε -equation requires both $m(t)$ and $\varepsilon_2(t)$ as external forcing, weakening the physics constraint, and ε_2 has no native ODE in the closed-moment system. Coefficient recovery degrades correspondingly. This validates m as the natural starting observable for the framework.

Monte Carlo is the natural ground truth, but it is also useful as a cost baseline. Generating one $T = 100$ trajectory at $N = 256$ spins, averaged over 8 realizations, requires on the order of seconds in the JIT-compiled simulator. PINN inference, by contrast, requires a single forward pass through a $\sim 13,000$ -parameter MLP, which is several orders of magnitude faster. The PINN therefore acts as a fast surrogate once trained, with the additional benefit that its predictions are constrained to satisfy the closed-moment ODE.

The chosen pipeline has three limitations worth noting. First, the independent-spins closure $\varepsilon_2 \approx m^2$ used inside the m -equation residual is exact only at infinite temperature; for finite β it introduces a small systematic bias whose magnitude grows in the strongly coupled system. Second, the framework

is condition-specific: each (β, h, J) requires its own training run. A multi-condition extension that learns a shared core is left for future work. Third, the underlying chain is one-dimensional, where the closed-moment expansion is particularly clean; the same approach in two or three dimensions would require either higher-order moment closures or a return to the full configurational state.

VI. DISCUSSION AND CONCLUSION

I have presented a PINN framework that learns the dynamics of an Ising-like Zimm-Bragg chain by embedding the closed-moment Glauber ODE directly in the training loss of a small multilayer perceptron. The chosen pipeline trains only on the helicity $\hat{m}(t)$ and recovers the full set of three observables $(\hat{m}(t), \hat{\varepsilon}, \hat{\varepsilon}_2)$ together with the underlying Glauber coefficients (a_0, a_1, a_2) . The two correlators are derived analytically from the trained $\hat{m}(t)$, with no additional networks or supervision.

Across a 100-condition grid, the network reproduces the Monte Carlo trajectories at MSE on the order of 10^{-7} on the helicity and matches the analytical Glauber forward map to within $\sim 10^{-3}$ on the recovered coefficients. A head-to-head benchmark against four alternative PINN designs (separate-channel, inverse, 3-channel, per-run) confirms that the analytical post-hoc reduction is the most parameter-efficient and most stable of the candidates.

Future work includes extension to a multi-condition shared-core architecture, application to two and three-dimensional Ising chains where the closed-moment expansion does not terminate at low order, and direct application to experimental polypeptide melting data, where the model could serve as a fast inverse-problem solver for the underlying Zimm-Bragg parameters.

APPENDIX A

DERIVATION OF THE CLOSED-MOMENT GLAUBER ODES

This appendix derives the system (2)-(3) from the microscopic master equation. The argument follows the line of reasoning of Glauber [3] and the helix-coil kinetics derivation of Allahverdyan et al. [4], specialized to the heat-bath rule used in the simulator and made compact for the one-dimensional chain.

Let $P(\boldsymbol{\sigma}, t)$ denote the probability of finding the chain in configuration $\boldsymbol{\sigma} = (\sigma_1, \dots, \sigma_N)$ at time t . Single-spin-flip Glauber dynamics evolve P according to

$$\frac{\partial P(\boldsymbol{\sigma}, t)}{\partial t} = \sum_{i=1}^N [w_i(\boldsymbol{\sigma}^{(i)})P(\boldsymbol{\sigma}^{(i)}, t) - w_i(\boldsymbol{\sigma})P(\boldsymbol{\sigma}, t)], \quad (11)$$

where $\boldsymbol{\sigma}^{(i)}$ is the configuration with spin i flipped and $w_i(\boldsymbol{\sigma})$ is the rate at which σ_i flips given the current configuration. The heat-bath rule $p(\sigma_i \rightarrow +1) = \frac{1}{2}(1 + \tanh \beta f_i)$ corresponds to a flip rate

$$w_i(\boldsymbol{\sigma}) = \frac{1}{2} [1 - \sigma_i \tanh \beta f_i], \quad (12)$$

with local field $f_i = h + J(\sigma_{i-1} + \sigma_{i+1})$.

Multiplying (11) by σ_k and summing over all configurations gives the standard result that, for any single-spin observable,

$$\frac{d}{dt} \langle \sigma_k \rangle = -2 \langle \sigma_k w_k(\boldsymbol{\sigma}) \rangle. \quad (13)$$

Substituting (12) into (13),

$$\frac{d}{dt} \langle \sigma_k \rangle = -\langle \sigma_k \rangle + \langle \tanh \beta f_k \rangle. \quad (14)$$

The remaining task is to evaluate $\langle \tanh \beta f_k \rangle$.

The local field f_k depends on $(\sigma_{k-1}, \sigma_{k+1})$, which takes only four values, but $\sigma_{k-1} + \sigma_{k+1}$ takes only three: $+2, 0, -2$. Writing the indicator algebra $\delta_{S=+2} = \frac{1}{4}(1 + \sigma_{k-1})(1 + \sigma_{k+1})$ etc. and substituting,

$$\begin{aligned} \tanh \beta f_k &= A + B(\sigma_{k-1} + \sigma_{k+1}) + C \sigma_{k-1} \sigma_{k+1}, \\ A &= \frac{1}{4} [\tanh \beta (h + 2J) + 2 \tanh \beta h + \tanh \beta (h - 2J)], \\ B &= \frac{1}{4} [\tanh \beta (h + 2J) - \tanh \beta (h - 2J)], \\ C &= \frac{1}{4} [\tanh \beta (h + 2J) - 2 \tanh \beta h + \tanh \beta (h - 2J)]. \end{aligned} \quad (15)$$

The three coefficients $(2A, 2B, 2C)$ are exactly (a_0, a_1, a_2) in (4). Substituting (15) into (14) and chain-averaging yields

$$\dot{m} = -m + a_0/2 + a_1 m + (a_2/2) \langle \sigma_{k-1} \sigma_{k+1} \rangle, \quad (16)$$

which is (2) after recognizing $\varepsilon_2 = \langle \sigma_{k-1} \sigma_{k+1} \rangle$ as the next-nearest-neighbor correlator. The companion equation (3) is derived analogously, starting from $d\langle \sigma_k \sigma_{k+1} \rangle / dt$.

APPENDIX B

DERIVATION OF THE PHYSICS LOSS

This appendix records how the physics term $\mathcal{L}_{\text{phys}}$ in (7) is obtained from the closed-moment ODE.

The closed m-equation (5) can be written in residual form by moving every term to one side:

$$R_m(t) := \dot{m}(t) + (1 - a_1)m(t) - \frac{a_0}{2} - \frac{a_2}{2} m(t)^2. \quad (17)$$

A function $m(\cdot)$ is a solution of (5) on $[0, T]$ if and only if $R_m(t) \equiv 0$ for all $t \in [0, T]$. The PINN replaces this pointwise zero condition with a sampled mean-square objective, which is the standard PINN construction [5] and is the residual definition shown in (7) when $\hat{m}$ replaces the unknown m and the trainable scalars (a_0, a_1, a_2) replace the analytical Glauber coefficients.

The time derivative $\dot{m}(t_{c,j})$ in $\text{Res}_m(t_{c,j})$ is obtained by backpropagating through the network with respect to its scalar input t , evaluated at the collocation point $t_{c,j}$. There are two implementation details.

First, the smooth Softplus activation in the MLP is what makes this derivative useful: a ReLU network has a piecewise-constant first derivative, whose samples are visited by the residual loss at each batch and produce a non-smooth model that defeats the L-BFGS line search. Second, the collocation points $t_{c,j}$ are resampled at each L-BFGS outer iteration; this is technically a Monte Carlo estimator of the integral of R_m^2 , and resampling reduces its variance.

The residual (17) hides one closure step: the exact m-equation (2) contains $\varepsilon_2(t)$, which has been replaced by $m(t)^2$.

This is the independent-spins approximation $\langle \sigma_{k-1} \sigma_{k+1} \rangle \approx \langle \sigma_k \rangle^2$, which is exact at $\beta = 0$ and is asymptotic at high temperature; its leading-order correction is $\mathcal{O}(\beta J)$, which sets the lower bound on the achievable residual error. This approximation enters only inside the residual that drives training; the analytical post-hoc step in Appendix C recovers $\varepsilon_2(t)$ without it.

Combining the supervised data term (6) and the residual term (7),

$$\mathcal{L}_{\text{total}} = \underbrace{\frac{1}{N_d} \sum_{i=1}^{N_d} (f_\theta(t_i) - \bar{m}_i)^2}_{\mathcal{L}_{\text{data}}} + \underbrace{\frac{1}{N_c} \sum_{j=1}^{N_c} R_m(t_{c,j})^2}_{\mathcal{L}_{\text{phys}}}, \quad (18)$$

where $\bar{m}_i = \bar{m}(t)[t_i]$ is the chain-and-ensemble averaged Monte Carlo helicity at sampling time t_i . No separate weighting hyperparameter is introduced between the two terms: in this setup both $\mathcal{L}_{\text{data}}$ and $\mathcal{L}_{\text{phys}}$ sit at comparable magnitudes by the end of the Adam phase, and L-BFGS is run on the unweighted sum.

APPENDIX C

RECOVERING $\varepsilon(t)$ AND $\varepsilon_2(t)$ FROM $m(t)$

The PINN does not directly target $\varepsilon(t)$, so it is estimated from $\hat{m}(t)$ alone. Under the independent-spins approximation,

$$\langle \sigma_k \sigma_{k+1} \rangle \approx \langle \sigma_k \rangle \langle \sigma_{k+1} \rangle = m^2, \quad (19)$$

which gives the estimator $\hat{\varepsilon}(t) = \hat{m}(t)^2$ used in (9). For the high and intermediate-temperature conditions on the parameter grid it is accurate to $\text{MSE} \sim 10^{-5}$. Strictly speaking it is the only step in the post-hoc reduction that is not closure-free, and so it is the dominant systematic error in $\hat{\varepsilon}(t)$.

By contrast, $\varepsilon_2(t)$ is recovered *without* any closure assumption. Returning to the exact m-equation (2),

$$\dot{m} + (1 - a_1) m - \frac{a_0}{2} = \frac{a_2}{2} \varepsilon_2, \quad (20)$$

which is solved algebraically for ε_2 :

$$\varepsilon_2(t) = \frac{2}{a_2} \left[\dot{m}(t) + (1 - a_1) m(t) - \frac{a_0}{2} \right]. \quad (21)$$

Replacing the analytical m and coefficients with their PINN estimates yields (10). Equation (21) contains no approximation: given any trajectory $m(t)$ that satisfies (2) together with the correct scalars, this expression returns the exact $\varepsilon_2(t)$. Its accuracy in practice is therefore determined entirely by how faithfully $\hat{m}(t)$ tracks $m(t)$ and how cleanly the trainable scalars match the analytical Glauber coefficients in (4).

Two numerical hazards are worth flagging. First, the inversion (21) divides by a_2 , which vanishes at $J = 0$ and is small in the high-temperature system more generally. Conditions with $|a_2| < 10^{-3}$ are therefore numerically ill-conditioned, but they are also exactly the conditions under which the m-equation is independent of ε_2 , so the inversion is useful only away from this boundary. Second, the time derivative $\dot{m}(t)$ in (10) is evaluated by automatic differentiation of the trained network, and inherits any high-frequency noise of the trained

$\hat{m}(t)$. In practice this is not an issue: the Softplus MLP is smooth by construction, and the $\hat{m}(t)$ trajectories obtained after L-BFGS convergence have irregularities in $\dot{m}(t)$ well below the closure-bias.

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